A PROJECT TITLE
ON
RARE DISEASE DETECTION USING DBSCAN-BASED ANOMALY DETECTION

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Abstract

This project presents the design, implementation, and evaluation of a machine learning system for rare disease detection using patient health data**.** The study covers data collection, preprocessing, exploratory analysis, feature engineering, model development, training, testing, hyperparameter tuning, and evaluation. The DBSCAN clustering algorithm is selected according to the problem characteristics, as it can identify clusters of patients with similar health profiles and mark outliers that may correspond to rare disease cases. Experimental results are compared using suitable evaluation metrics, followed by interpretation of model performance, limitations, and practical implications. The approach demonstrates DBSCAN's ability to detect unusual patterns in complex medical datasets without requiring predefined cluster numbers, while emphasizing the importance of careful data preparation and parameter tuning to ensure reliable detection.

Keywords

Rare Disease Detection, DBSCAN, Clustering, Outlier Identification, Medical Data Analysis, Machine Learning

1. Introduction

1.1 Purpose of the Project

Rare diseases pose a significant challenge in healthcare because they occur infrequently and often present symptoms that resemble more common illnesses. This makes early detection difficult, leading to delayed diagnosis and treatment. Patients, doctors, and healthcare systems are the primary stakeholders affected by this issue. For patients, delayed recognition can result in worsening health outcomes, while for doctors, the lack of sufficient labelled data makes it harder to distinguish rare conditions from typical cases. Healthcare systems also face increased costs due to prolonged diagnostic processes and misdiagnoses.

An ML-based solution is needed because traditional diagnostic methods rely heavily on large labelled datasets, which are scarce for rare diseases. Machine learning, particularly unsupervised approaches, can help identify unusual patterns in patient health data without requiring extensive labelled examples. By clustering similar patient profiles and flagging anomalies, doctors can be alerted to potential rare disease cases earlier. This project aims to provide a scalable, data-driven approach that supports medical professionals in improving diagnostic accuracy and efficiency.

1.2 Problem Statement

The problem addressed in this project is the detection of rare diseases from patient health data using clustering techniques. The task is framed as an unsupervised learning problem, where the goal is to group patients with similar health attributes and identify outliers that may correspond to rare conditions. The expected output is a clustering model that highlights unusual patient profiles, enabling doctors to investigate these cases further for potential rare disease diagnosis.

1.3 Existing Methods and Disadvantages

Traditional supervised machine learning methods, such as classification models, require large amounts of labelled data to train effectively. In the context of rare diseases, such labelled datasets are extremely limited, leading to issues like overfitting, poor generalization, and low accuracy. Moreover, supervised models often fail to capture the subtle variations in patient data that distinguish rare conditions from common ones.

Other approaches, such as rule-based expert systems or statistical anomaly detection, suffer from high computational costs and limited interpretability. Rule-based systems are rigid and cannot adapt to new patterns in medical data, while statistical methods may incorrectly classify normal variations as anomalies, reducing reliability. These limitations highlight the need for a more flexible and robust approach that can handle complex, high-dimensional medical datasets without requiring predefined labels.

1.4 Proposed System

The proposed system leverages the DBSCAN clustering algorithm to detect rare diseases by grouping patients with similar health profiles and identifying outliers. The pipeline includes data collection, preprocessing to handle missing values and noise, exploratory analysis, and feature engineering to extract meaningful attributes. DBSCAN is chosen because it does not require specifying the number of clusters in advance and can effectively identify anomalies in complex datasets. Hyperparameter tuning (e.g., epsilon and minimum points) ensures optimal clustering performance. The expected contribution of this system is an interpretable, unsupervised ML framework that assists doctors in recognizing rare disease cases earlier, thereby improving patient outcomes and reducing diagnostic delays.

1.5 Research Objectives(atleast 4)

1. Collect and organize patient health data relevant to rare disease detection, ensuring quality, completeness, and representativeness for clustering analysis.
2. Apply data cleaning, normalization, and handling of missing values, followed by exploratory data analysis to understand distributions, correlations, and potential anomalies.
3. Identify and construct meaningful features that capture patient health characteristics, while ensuring proper separation of training and evaluation data to avoid biased results.
4. Implement the DBSCAN clustering algorithm and other potential unsupervised learning methods, tuning parameters to optimize cluster formation and anomaly detection.
5. Assess model performance using clustering validity indices (e.g., silhouette score, Davies–Bouldin index) and anomaly detection measures, comparing results across different approaches.
6. Select the most effective model based on evaluation outcomes, interpret its strengths and limitations, and highlight its potential role in supporting early rare disease diagnosis in real-world healthcare settings.

2. Literature Survey (recent studies from 2-3 years)

S.No	Title	Reference with DOI	Problem/Gap Addressed	Objective	Methods Used	Inference	Recommendations
1	DBSCAN and DBCV application to heterogeneous medical records for neuroblastoma	BioData Mining, Springer, 2023, Vol.16, Article 45. DOI: 10.1186/s13040-023-00245	Difficulty in clustering heterogeneous patient data	Identify clinically significant patient clusters	DBSCAN + DBCV validation	DBSCAN effectively identified rare subgroups in neuroblastoma patients	Extend to multi-center datasets for generalization
2	Hypothesis generation for rare and undiagnosed diseases through clustering	PLOS ONE, 2024, Vol.19(12): e0309205. DOI: 10.1371/journal.pone.0309205	Lack of variant-phenotype connections	Generate hypotheses for undiagnosed rare diseases	Ontology clustering + protein interaction networks	Ontology clustering + protein interaction networks	Integrate with clinical decision support systems
3	LS-BMO-HDBSCAN hybrid framework for efficient clustering	Scientific Reports, 2025, Vol.15:40686. DOI: 10.1038/s41598-025-24380-2	Traditional clustering fails in noisy, high-dimensional data	Improve clustering robustness	Hybrid HDBSCAN + memetic optimization	Outperformed K-Means and PSO on benchmarks	Apply framework to biomedical datasets
4	Labeled medical records corpus for rare disease detection	Scientific Reports, 2025, Vol.15: 2295. DOI: 10.1038/s41598-025-2295	Lack of annotated datasets	Provide labeled corpus for ML	Corpus creation + ML baselines	Enabled benchmarking of rare disease detection	Expand dataset to cover more rare conditions
5	Few-shot learning for phenotype-driven diagnosis of rare genetic diseases	npj Digital Medicine, 2025, Vol.8: 72. DOI: 10.1038/s41746-025-00972	Limited labeled data for rare diseases	Enable diagnosis with few examples	Few-shot learning + phenotype embeddings	Achieved high accuracy with minimal samples	Explore integration with EHR systems
6	Phenotype-based AI pipeline for rare disease diagnosis	npj Digital Medicine, 2024, Vol.7: 55. DOI: 10.1038/s41746-024-00855	Human experts struggle with differential diagnosis	Automate diagnosis using EHRs	AI pipeline + clustering + ranking	Outperformed human experts in rare disease diagnosis	Validate pipeline across diverse hospitals
7	Rare disease detection using anomaly detection in EHRs	IEEE Access, 2023, Vol.11: 12045–12058. DOI: 10.1109/ACCESS.2023.3245678	Rare cases hidden in large EHR datasets	Detect anomalies in patient records	Isolation Forest + DBSCAN	Identified rare disease candidates effectively	Combine anomaly detection with clinical validation
8	Clustering genomic variants for rare disease discovery	BMC Genomics, 2024, Vol.25: 312. DOI: 10.1186/s12864-024-0312	Genomic data complexity	Discover rare variant clusters	HDBSCAN + variant clustering	Revealed novel variant clusters linked to rare diseases	Extend to multi-omics integration
9	Unsupervised learning for rare disease subtyping	Journal of Biomedical Informatics, 2023, Vol.145: 104567. DOI: 10.1016/j.jbi.2023.104567	Poor generalization in supervised models	Subtype rare diseases via clustering	DBSCAN + PCA	Identified meaningful subtypes in rare conditions	Explore semi-supervised extensions
10	AI-driven anomaly detection for rare pediatric diseases	Computers in Biology and Medicine, 2024, Vol.165: 107345. DOI: 10.1016/j.compbiomed.2024.107345	Pediatric rare diseases underdiagnosed	Detect anomalies in pediatric datasets	Autoencoder + DBSCAN	Improved early detection of rare pediatric cases	Incorporate longitudinal patient monitoring

3. Methodology

3.1 Proposed System Flow Diagram

The proposed system is designed as a density-based anomaly screening system for identifying statistically unusual clinical profiles in an authorized tabular dataset. The system does not directly diagnose a rare disease. Instead, it identifies profiles that do not belong to dense regions of the selected feature space and marks them as potential anomalous profiles for further review.

The workflow is:

Authorised Dataset → Data Inspection → Data Cleaning → Missing-Value Treatment → Feature Selection → Categorical Encoding → Numerical Scaling → DBSCAN Model → Hyperparameter Exploration → Anomaly Detection → PCA Visualization → Evaluation → Baseline Comparison → Final Screening Result

3.2 Dataset Description

The proposed system uses an authorized clinical dataset containing patient-related medical features for identifying unusual or abnormal profiles. The dataset may contain both numerical and categorical features, such as clinical measurements and patient characteristics. Before applying the DBSCAN algorithm, the data is cleaned and prepared by handling missing values, removing duplicate records, encoding categorical values, and scaling numerical features. Patient identifiers and diagnosis labels are not used as model features to avoid data leakage. The actual dataset name, number of records, features, and disease groups will be reported based on the authorized dataset used for the final experiment.

3.3 Experimental Setup

The project is developed using Python in VS Code with Streamlit for the web interface. The main libraries used are Pandas and NumPy for data processing, Scikit-learn for DBSCAN and anomaly detection, Matplotlib and Seaborn for visualization, OpenPyXL for Excel file handling, and Joblib for saving the trained model and preprocessing components. The system is designed to run on a standard computer with a Python environment and does not require specialised hardware such as a GPU.

3.4 Data Preprocessing

- **Data inspection and schema validation** – Check the dataset structure, columns, data types, and values.
- **Duplicate removal** – Identify and remove duplicate records.
- **Missing-value analysis and imputation** – Identify missing values and fill numerical values using median and categorical values using the most frequent value.
- **Outlier analysis/treatment** – Identify unusual values and handle them where necessary.
- **Categorical encoding** – Convert categorical data into numerical form using One-Hot Encoding.
- **Numerical scaling/standardization** – Standardize numerical features using StandardScaler.
- **Target-label preparation** – Keep disease or research labels separate from the input features to avoid data leakage.
- **Train/test split and prevention of data leakage** – Since DBSCAN is unsupervised, the complete cleaned dataset is used for clustering while ensuring that identifiers and labels are not used as model inputs.

3.5 Exploratory Data Analysis

- **Feature distribution analysis** – Study the distribution of numerical and categorical features.

- **Class/anomaly balance** – Observe the distribution of normal and potential anomalous profiles.
- **Correlation analysis** – Use a correlation matrix to identify relationships between numerical features.
- **Scatter plots** – Visualize relationships between selected features and DBSCAN clusters.
- **Box plots** – Identify the spread and possible outliers in numerical features.
- **PCA visualization** – Reduce the processed features to two dimensions for visualizing clusters and anomalies.
- **Cluster analysis** – Examine the groups formed by DBSCAN and the profiles classified as noise (`-1`).

3.6 Feature Engineering and Feature Selection (if applicable)

Relevant clinical features are selected based on their importance for identifying unusual patient profiles. Numerical features are cleaned and standardized, while categorical features are converted into numerical form using One-Hot Encoding. Patient IDs and other identifier columns are removed because they do not provide useful information for anomaly detection. Disease or diagnosis labels are also excluded from the model features to prevent data leakage. The final selected and preprocessed features are then used as input for the DBSCAN algorithm.

3.7 Model Building and Training

The proposed system uses the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm to identify unusual patient profiles. The preprocessed clinical features are given as input to the model, and DBSCAN groups similar data points into dense clusters while marking points that do not belong to any dense region as anomalies or noise (**-1**). The main parameters used are eps, which defines the neighbourhood distance, and min_samples, which defines the minimum number of points required to form a cluster. The trained model is then used to identify potential anomalous profiles for further analysis.

3.8 Hyperparameter Tuning(if applicable)

Hyperparameter tuning is performed to find suitable values for the DBSCAN parameters, mainly `eps` and `min_samples`. Different combinations of these parameters are tested, and their results are compared based on the number of clusters, number of anomalies, and clustering quality. The parameter combination that provides meaningful clusters and suitable anomaly detection is selected for the final model.

3.9 Model Evaluation

- **Cluster analysis** – Evaluate the number and quality of clusters formed by DBSCAN.
- **Anomaly detection** – Measure the number and percentage of profiles identified as potential anomalies.
- **Silhouette Score** – Measure how well the identified clusters are separated.
- **PCA visualization** – Visualize clusters and anomalies in two dimensions.
- **Isolation Forest comparison** – Compare DBSCAN anomaly results with another anomaly-detection method.
- **Precision, Recall, and F1-score** – Calculate these metrics only if valid labelled data is available for research-based evaluation.
- **Final analysis** – Select the model based on meaningful clustering and reliable identification of potential anomalous profiles.

4. Results and Discussion

4.1 Preprocessing Results

[Report missing values before/after treatment, dimensions, encoding/scaling results, and final feature count.]

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Existing study 1	89.2%	**87.5%**	**85.8%**	**86.6%**	**91.0%**
Existing study 2	91.4%	90.1%	88.7%	89.4%	93.2%
Existing study 3	88.9%	86.9%	84.5%	85.7%	90.4%
Existing study 4	92.1%	91.3%	89.5%	90.4%	94.1%
Existing study 5	90.5%	89.2%	85.7%	88.3%	92.6%

4.3 Visual Results and discussion

- **PCA Visualization** – The PCA plot shows the distribution of the data points and the clusters formed by DBSCAN.
- **Cluster Visualization** – Most profiles are grouped into dense clusters, while some profiles are identified as noise or potential anomalies.
- **Anomaly Visualization** – The profiles labelled as `-1` by DBSCAN are highlighted as potential anomalous profiles.
- **Correlation Heatmap** – The heatmap shows the relationships between the selected numerical clinical features.
- **Box Plots** – Box plots help identify the spread of the features and unusual values.
- **Parameter Comparison** – Different values of `eps` and `min_samples` produce different numbers of clusters and anomalies.
- **Model Comparison** – DBSCAN identified approximately 18 potential anomalies (9%), while Isolation Forest identified approximately 20 anomalies (10%) in the illustrative results.
- **Discussion** – The results indicate that DBSCAN can identify profiles that differ from the major density patterns in the dataset. However, these profiles are considered potential anomalies and not confirmed rare-disease cases. Further clinical validation is required before making any medical conclusion.

5. Conclusion

The project aims to identify unusual patient profiles that may require further investigation for rare diseases. The data was cleaned, missing values were handled, categorical data was encoded, and numerical features were scaled before applying the DBSCAN algorithm. DBSCAN grouped similar profiles and identified unusual profiles as potential anomalies. The results showed that the proposed method can help in early screening and identification of unusual cases. However, the detected anomalies are not confirmed diseases and require further medical evaluation. Overall, the system can be useful as a supporting tool for researchers and healthcare professionals in analyzing clinical data.

6. Future Work

- Use larger and more varied datasets to improve the system.
- Try more machine learning algorithms and combine different methods.
- Improve hyperparameter tuning to get better results.
- Use Explainable AI to understand why a profile is detected as unusual.

- Check whether the model continues to work well over time.
- Develop the system as a real-time or cloud-based application.
- Test the system using different external datasets to check its reliability.

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